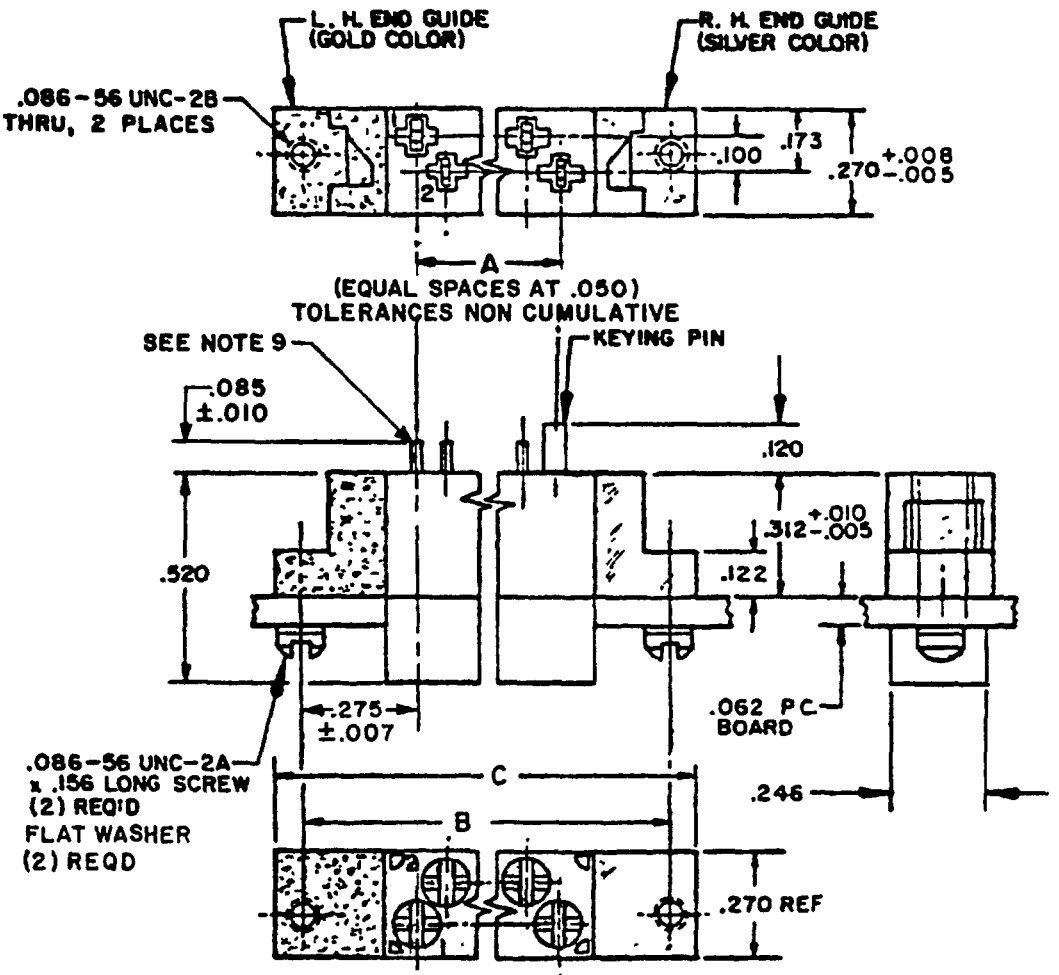
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG STRAIGHT-THRU, HERMAPHRODITIC CONTACT, CRIMP REMOVABLE, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by the Communications Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



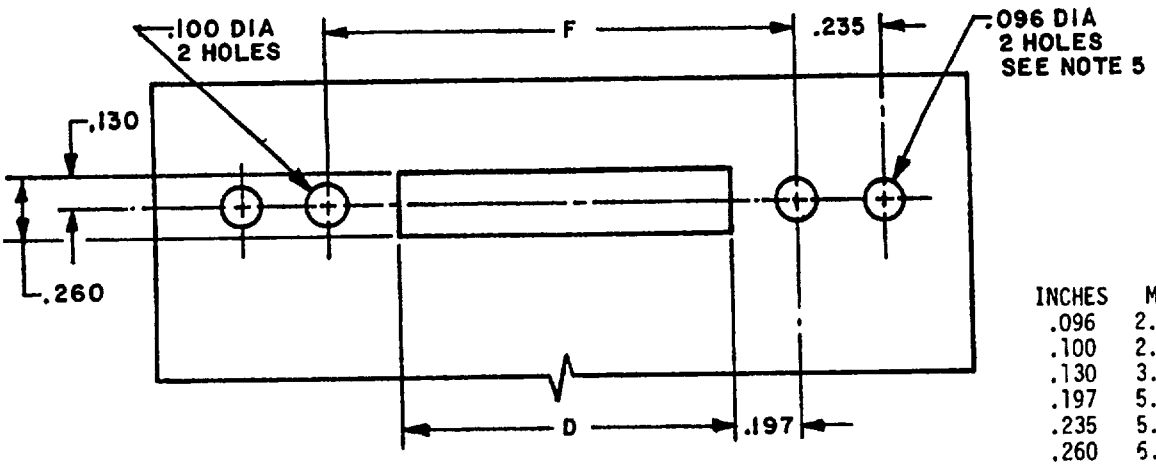
INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.085	2.16	.173	4.39
.007	.18	.086	2.18	.246	6.25
.008	.20	.100	2 54	.270	6.86
.010	.25	.120	3 05	.275	6.98
.050	1.27	.122	3.10	.312	7.92
.062	1.57	.156	3.96	.520	13.21

FIGURE 1. Connector, plug (.100 spacing).

Ⓓ denotes changes

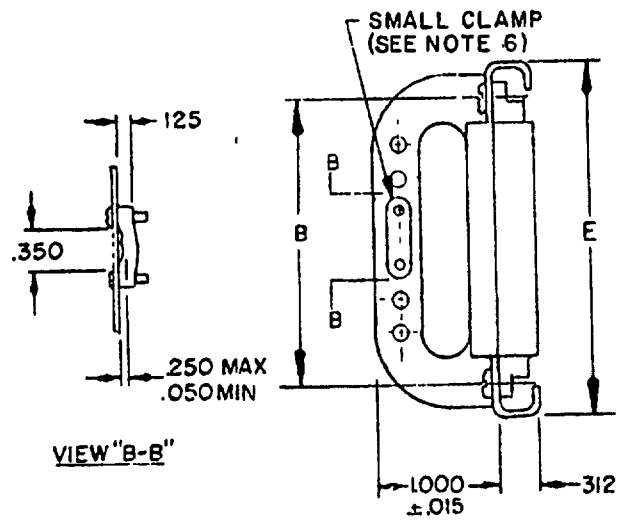
FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.



INCHES	MM
.096	2.44
.100	2.54
.130	3.30
.197	5.00
.235	5.97
.260	6.60

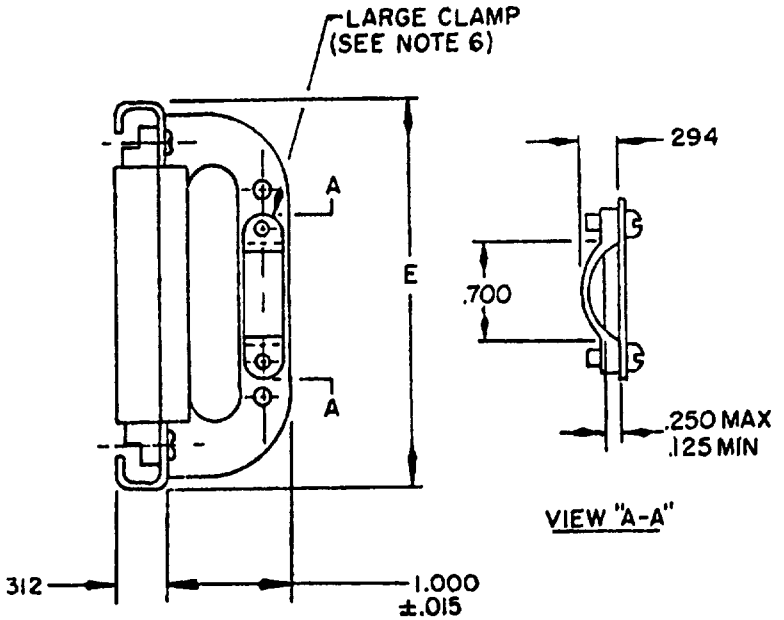
FIGURE 2. Typical cutout for plug for nonconductive chassis.



INCHES	MM
.015	.38
.050	1.27
.125	3.18
.250	6.35
.312	7.92
.350	8.89
1.000	25.40

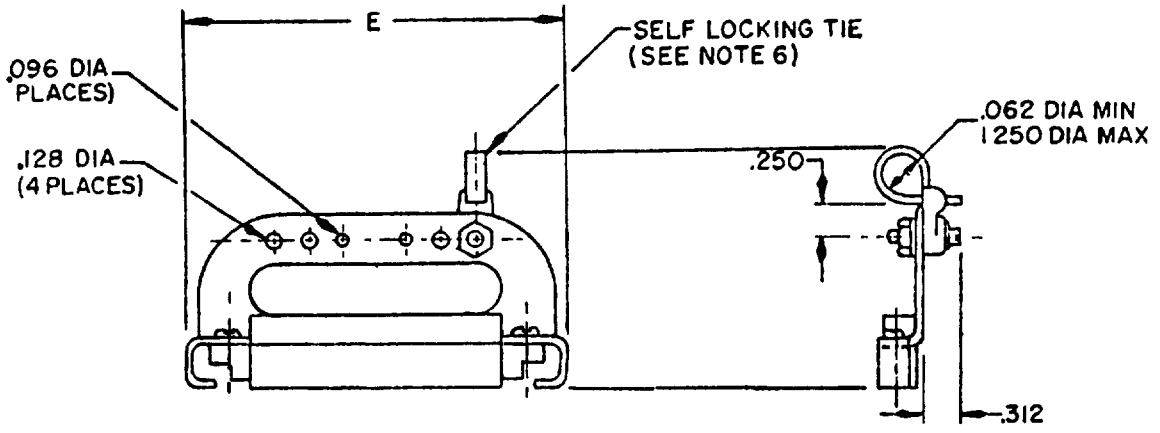
FIGURE 3. Mounting bracket assembly with small clamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.



INCHES	MM
.015	.38
.125	3.18
.250	6.35
.294	7.47
.312	7.92
.700	17.78
1.000	25.40

FIGURE 4. Mounting bracket assembly with large clamp.



INCHES	MM
.062	1.57
.096	2.44
.128	3.25
.250	6.35
.312	7.92
1.250	31.75

FIGURE 5. Mounting bracket assembly with self locking tie.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. When keying, mounting bracket or mounting bracket and keying is required, refer to the applicable dash and figure number in table I.
5. When bracket assembly (see figures 3, 4, or 5) to chassis mounting is required, it is suggested that support pins per MIL-C-55302/107 be used.
6. Mounting brackets assembled with the insulator will have the clamping hardware shipped loose.
7. Contacts (class A per MIL-C-55302/97-01) are shipped loose in a plastic bag with each connector unless otherwise specified. When contacts are furnished separately, the letters "WO" shall appear immediately after the military part number marking on the connector (i.e., M55302/102-02 WO).
8. For direct government procurements connectors shall be furnished with contacts in a plastic bag with each connector.
9. With connectors fully engaged the mating contacts shall have a minimum overlap distance of .062. As a minimum the contacts shall exhibit a wiping action through this distance.

TABLE I. Dash numbers and dimensions.

Dash number 1/	No. of contacts	Fig. no.	*Dimensions (inches) A	B ±.010	C Ref.	D	E Ref.	F
01 02 03 04	18	1 3 4 5	8.50 (21.59)	1.400 (35.56)	1.540 (39.12)	1.006 (25.55)	1.990 (50.55)	1.400 (35.36)
05 06 07 08	30	1 3 4 5	1.450 (36.83)	2.000 (50.80)	2.140 (54.36)	1.606 (40.79)	2.590 (65.79)	2.000 (50.80)
09 10 11 12	36	1 3 4 5	1.750 (44.45)	2.300 (58.42)	2.440 (61.98)	1.906 (48.41)	2.890 (73.41)	2.300 (58.42)
13 14 15 16	42	1 3 4 5	2.050 (52.07)	2.600 (66.04)	2.740 (69.60)	2.206 (56.03)	3.190 (81.03)	2.600 (66.04)
17 18 19 20	54	1 3 4 5	2.650 (67.31)	3.200 (81.28)	3.340 (84.84)	2.806 (71.27)	3.790 (96.27)	3.200 (81.28)
21 22 23 24	72	1 3 4 5	3.550 (90.17)	4.100 (104.14)	4.240 (107.70)	3.706 (94.13)	4.690 (119.13)	4.100 (104.14)

1/ See requirements for complete part number when keying pin is required.
* Metrics are in parentheses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figures 1 through 5 and table I.

Material and finish:

Insulator. MIL-M-14, type SDG-F.

D Bracket, mounting: Cold rolled steel, 1/4 HD, per QQ-S-698; cadmium plated per QQ-P-416, class 3, type II.

End guides: Left hand, sintered brass 90/10, clear chromate per MIL-C-5541, class 1A. Right hand, sintered brass 90/10, nickel plate 300 microinches per MIL-F-14072.

Clamps, large and small: Aluminum alloy 380, per QQ-A-591, cadmium plated per QQ-P-416, class 3, type II.

Keying pin: Aluminum alloy 2011-T3 with clear chromate per MIL-C-5541, class 1A.

Self-locking tie: Nylon 6.6.

Mounting screw: Brass, cadmium plated, gold chromate finish.

Flat washer: Stainless steel.

Contact identification: Contact locations are identified in numerical sequence (see figure 1).

Mating and unmating: The maximum insertion force, in pounds, shall not exceed a value equal to .5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of .11 times the number of contacts and shall not exceed the measured insertion force.

Contact retention: 6.0 pounds minimum after five insertions and withdrawals from the same connector position.

Contact resistance: The contact resistance shall not exceed 20 milliohms.

Dry circuit: Connectors shall be tested in accordance with method 3002 of MIL-STD-1344. The contact resistance shall not exceed 20 milliohms.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms, 60 hertz.

High altitude: 300 volts rms, 60 hertz.

Current rating, maximum: 5 amperes.

Keying: When required, one of the following codes shall be added to the dash number:

"P" - Specifying location by contact number where keying pin shall be inserted; for example, M55302/104-01P17. (Keying pin inserted in place of contact number 17.)

"H" - Specifying location by contact number where contact shall be omitted for mating; for example, M55302/104-01H17. (Keying pin hole has contact number 17 omitted.)

Contact crimp removable (Class A or B): Shall conform to MIL-C-55302/97.

Contact removal tool: Shall conform to MIL-I-81969/4.

Contact insertion tool: Shall conform to MIL-I-81969/5.

Contact crimp tool: Shall conform to MIL-C-22520/17-02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

Mating receptacle: Shall conform to MIL-C-55302/52, MIL-C-55302/99, or MIL-C-55302/102.

Part number: M55302/104-(dash number from table I).

Custodians:
Army - CR

Review activities:
Army - AT, AV, MI
DLA - ES

User activities:
Army - AR, ME

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-A228-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

HQ, CECOM
ATTN: DRSEL-ED-TO
Ft Monmouth, N.J. 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://... Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.